

Project Structure

SeleniumAutomationProjectusingExcelandProperties

|— ConfigReader.java

|— ExcelReader.java

|— LoginBL.java

|— RegisterBL.java

|— AddUserBL.java

|— TestExecution.java

|— data.properties

|— Testdata.xlsx

Apache POI Jars Required

poi

poi-ooxml

commons-collections4 // optional

xmlbeans // optional

data.properties

browsername= Chrome

loginurl= https://javabykiran.com/liveproject/index.html

registerurl= https://javabykiran.com/liveproject/pages/examples/register.html

adduserurl= https://javabykiran.com/liveproject/pages/examples/add_user.html

Testdata.xlsx

Sheet 1: Login

Scenario	Email	Password
Positive	kiran@gmail.com	123456
Negative	abc@gmail.com	wrong123

Sheet 2: Register

Scenario	Name	Mobile	Email	Password
Positive	Rahul	9876543210	rahul@gmail.com	Rahul@123
Negative	(Blank)	123	abc	(Blank)

Sheet 3: AddUser

Scenario	Username	Mobile	Email	Course	Gender	State	Password
Positive	Amit	9876543210	amit@gmail.com	Java/J2EE	Male	Maharashtra	Amit@123
Negative	(Blank)	12	amit	(Blank)	(Blank)	(Blank)	(Blank)

1. ConfigReader.java

```
import java.io.FileInputStream;  
import java.util.Properties;
```

```
public class ConfigReader {
```

```
    Properties prop;
```

```
    public ConfigReader() {
```

```
        try {
```

```
            FileInputStream fis =  
                new FileInputStream("data.properties");
```

```
            prop = new Properties();
```

```
            prop.load(fis);
```

```
        } catch (Exception e) {
```

```
            e.printStackTrace();
```

```
        }  
    }
```

```
    public String getBrowser() {
```

```

    return prop.getProperty("browsername");
}

public String getLoginURL() {

    return prop.getProperty("loginurl");
}

public String getRegisterURL() {

    return prop.getProperty("registerurl");
}

public String getAddUserURL() {

    return prop.getProperty("adduserurl");
}
}

```

2. ExcelReader.java

```

import java.io.FileInputStream;
import org.apache.poi.ss.usermodel.*;

public class ExcelReader {

    public static String getData(String sheetName, int row, int column) {

        String value = "";

        try {

            FileInputStream fis = new FileInputStream("TestData.xlsx");

            Workbook wb = WorkbookFactory.create(fis);

            Sheet sheet = wb.getSheet(sheetName);

            value = sheet.getRow(row)
                .getCell(column)
                .getStringCellValue();

```

```
        wb.close();

    } catch (Exception e) {

        e.printStackTrace();
    }

    return value;
}

}
```

3. LoginBL.java

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class LoginBL {

    WebDriver driver;

    public LoginBL(WebDriver driver) {

        this.driver = driver;
    }

    public void login(String email, String password) {

        driver.findElement(By.id("email")).clear();
        driver.findElement(By.id("email")).sendKeys(email);

        driver.findElement(By.id("password")).clear();
        driver.findElement(By.id("password")).sendKeys(password);

        driver.findElement(By.xpath("//button")).click();

        System.out.println("Login Executed");
    }

}
```

4. RegisterBL.java

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class RegisterBL {

    WebDriver driver;

    public RegisterBL(WebDriver driver) {

        this.driver = driver;
    }

    public void register(String name,
                        String mobile,
                        String email,
                        String password) {

        driver.findElement(By.id("name")).clear();
        driver.findElement(By.id("name")).sendKeys(name);

        driver.findElement(By.id("mobile")).clear();
        driver.findElement(By.id("mobile")).sendKeys(mobile);

        driver.findElement(By.id("email")).clear();
        driver.findElement(By.id("email")).sendKeys(email);

        driver.findElement(By.id("password")).clear();
        driver.findElement(By.id("password")).sendKeys(password);

        driver.findElement(By.xpath("//button")).click();

        System.out.println("Registration Executed");
    }
}
```

5. AddUserBL.java

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.support.ui.Select;

public class AddUserBL {
```

WebDriver driver;

public AddUserBL(WebDriver driver) {

 this.driver = driver;
}

public void addUser(String username,
 String mobile,
 String email,
 String course,
 String gender,
 String state,
 String password) {

 driver.findElement(By.id("username")).clear();
 driver.findElement(By.id("username")).sendKeys(username);

 driver.findElement(By.id("mobile")).clear();
 driver.findElement(By.id("mobile")).sendKeys(mobile);

 driver.findElement(By.id("email")).clear();
 driver.findElement(By.id("email")).sendKeys(email);

 driver.findElement(By.id("course")).clear();
 driver.findElement(By.id("course")).sendKeys(course);

 if(gender.equalsIgnoreCase("Male")) {

 driver.findElement(By.id("Male")).click();

 } else if(gender.equalsIgnoreCase("Female")) {

 driver.findElement(By.id("Female")).click();

 }

 Select s = new Select(driver.findElement(By.id("state")));
 s.selectByVisibleText(state);

 driver.findElement(By.id("password")).clear();
 driver.findElement(By.id("password")).sendKeys(password);

```

        driver.findElement(By.xpath("//button")).click();

        System.out.println("Add User Executed");
    }

}

```

6. TestExecution.java

```

import java.time.Duration;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.edge.EdgeDriver;

public class TestExecution {

    public static void main(String[] args) throws Exception {

        ConfigReader config = new ConfigReader();

        WebDriver driver = null;

        if (config.getBrowser().equalsIgnoreCase("chrome")) {

            driver = new ChromeDriver();

        } else if (config.getBrowser().equalsIgnoreCase("edge")) {

            driver = new EdgeDriver();
        }

        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));

        // LOGIN Test

        driver.get(config.getLoginURL());

        LoginBL login = new LoginBL(driver);

        System.out.println("Positive Login");
    }
}

```

```
login.login(  
    ExcelReader.getData("Login", 1, 1),  
    ExcelReader.getData("Login", 1, 2));
```

```
Thread.sleep(3000);
```

```
driver.navigate().back();
```

```
System.out.println("Negative Login");
```

```
login.login(  
    ExcelReader.getData("Login", 2, 1),  
    ExcelReader.getData("Login", 2, 2));
```

```
Thread.sleep(3000);
```

```
// REGISTER Test
```

```
driver.get(config.getRegisterURL());
```

```
RegisterBL register = new RegisterBL(driver);
```

```
System.out.println("Positive Registration");
```

```
register.register(  
    ExcelReader.getData("Register", 1, 1),  
    ExcelReader.getData("Register", 1, 2),  
    ExcelReader.getData("Register", 1, 3),  
    ExcelReader.getData("Register", 1, 4));
```

```
Thread.sleep(3000);
```

```
driver.get(config.getRegisterURL());
```

```
System.out.println("Negative Registration");
```

```
register.register(  
    ExcelReader.getData("Register", 2, 1),  
    ExcelReader.getData("Register", 2, 2),  
    ExcelReader.getData("Register", 2, 3),  
    ExcelReader.getData("Register", 2, 4));
```

```
Thread.sleep(3000);
```



```
// ADD USER Test
```

```
driver.get(config.getAddUserURL());
```

```
AddUserBL addUser = new AddUserBL(driver);
```

```
System.out.println("Positive Add User");
```

```
addUser.addUser(  
    ExcelReader.getData("AddUser", 1, 1),  
    ExcelReader.getData("AddUser", 1, 2),  
    ExcelReader.getData("AddUser", 1, 3),  
    ExcelReader.getData("AddUser", 1, 4),  
    ExcelReader.getData("AddUser", 1, 5),  
    ExcelReader.getData("AddUser", 1, 6),  
    ExcelReader.getData("AddUser", 1, 7));
```

```
Thread.sleep(3000);
```

```
driver.get(config.getAddUserURL());
```

```
System.out.println("Negative Add User");
```

```
addUser.addUser(  
    ExcelReader.getData("AddUser", 2, 1),  
    ExcelReader.getData("AddUser", 2, 2),  
    ExcelReader.getData("AddUser", 2, 3),  
    ExcelReader.getData("AddUser", 2, 4),  
    ExcelReader.getData("AddUser", 2, 5),  
    ExcelReader.getData("AddUser", 2, 6),  
    ExcelReader.getData("AddUser", 2, 7));
```

```
Thread.sleep(3000);
```

```
// CLOSE BROWSER
```

```
System.out.println("Execution Completed Successfully");
```

```
driver.quit();
```

```
}
```

```
}
```